

Homework — 1.4.2 Data Structures

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 1.4.2 lesson. *SHOW ALL WORKING.*

Part A — This week's topic (~14 marks)

1. A **stack** and a **queue** are both initially empty.

(a) Complete the table to show the stack contents after each operation, and the value returned by each `pop()`. Write the contents **top first**; write — where no value is popped. [2] (AO2)

Operation	Stack after the operation (top first)	Value popped
<code>push(8)</code>		
<code>push(3)</code>		
<code>push(6)</code>		
<code>pop()</code>		
<code>pop()</code>		
<code>push(1)</code>		

(b) Complete the table for the queue. Write the contents **front first**; write — where no value is dequeued. [2] (AO2)

Operation	Queue after the operation (front first)	Value dequeued
<code>enqueue(4)</code>		
<code>enqueue(9)</code>		
<code>dequeue()</code>		
<code>enqueue(2)</code>		
<code>dequeue()</code>		

2. A **linked list** stores the values `5 → 12 → 20`. The table below shows how it is held in memory; the start pointer holds **100**. A new node containing **15** is inserted between 12 and 20, using the free node at address **400**. Complete the **After** table, filling in **every** Data and Next cell (write `null` for an empty pointer). The start pointer is unchanged. [3] (AO2)

Before the insertion:

Address	Data	Next
100	5	200
200	12	300
300	20	null
400	(free)	—

After the insertion:

Address	Data	Next
100		
200		
300		
400		

3. The following numbers are inserted, in this order, into an empty **binary search tree**: **40, 60, 25, 35, 10, 50**.

(a) Complete the table to show where each value is placed. The root row has been done for you. **[2]** (AO2)

Value	Parent node	Left or right child of its parent?
40	— (root)	—
60		
25		
35		
10		
50		

(b) Write the **in-order** traversal of this tree. **[1]** (AO2)

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4. Complete the table: write True or False for each statement, and write a correction for every false statement. **[2]** (AO1)

Statement	True / False	Correction if false
A queue is a LIFO structure.		
An in-order traversal of a binary search tree visits the values in ascending order.		
Inserting a value into a linked list means all the following nodes must be moved in memory.		
Pushing an item onto a stack that is already full causes a stack overflow.		

5. A hash table of size **10** uses the hash function $\text{index} = \text{key} \bmod 10$. The keys **23, 57, 33** are inserted in that order. State the index each key hashes to, identify any **collision**, and name **one** method of resolving it. **[2]** (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (1.2.1) Explain why an operating system uses a **LIFO stack** to handle **nested interrupts**. **[2]** (AO1)

7. (1.1.1) State the role of the **control bus** in a processor. **[1]** (AO1)

8. (1.3.2) A relational database links two tables using a **foreign key**. State what a foreign key is, and explain what **referential integrity** ensures. **[3]** (AO1)
